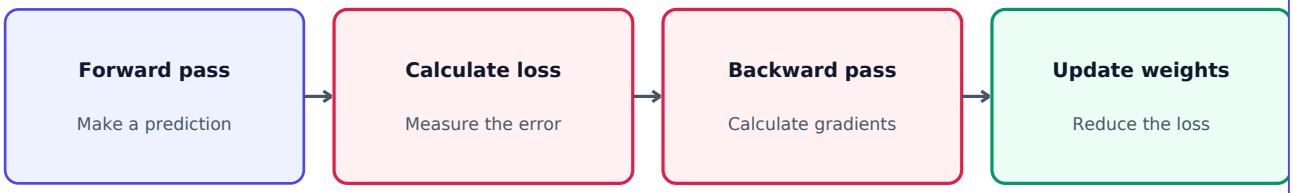
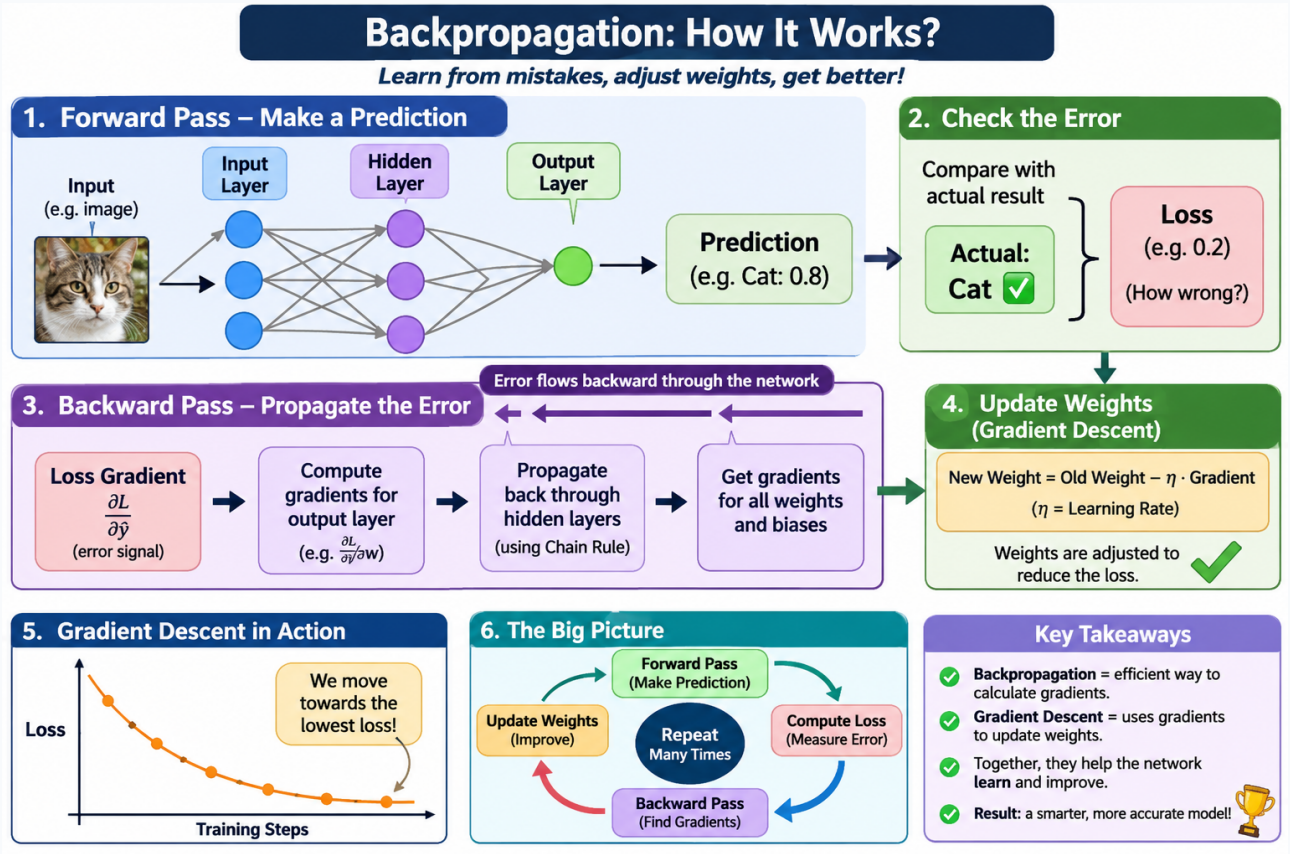
BACKPROPAGATION

How neural networks learn from their mistakes

The central idea

A neural network makes a prediction, measures how wrong it is, calculates how its weights contributed to the error, and adjusts those weights to improve.





A concise visual guide with history, prerequisites, working steps, applications and limitations.

1. Why is backpropagation so important?

Backpropagation became a major breakthrough because it gave neural networks an efficient way to calculate how their internal weights should change when a prediction is wrong.

Why it matters

- It made training networks with multiple layers much more practical.
- It allowed networks to learn useful patterns and features from examples.
- It became a central training method in modern deep learning.

Why is Geoffrey Hinton called the “Godfather of AI”?

Geoffrey Hinton made major contributions to neural networks and deep learning. In 2024, he shared the Nobel Prize in Physics with John Hopfield for foundational discoveries and inventions that enable machine learning with artificial neural networks.

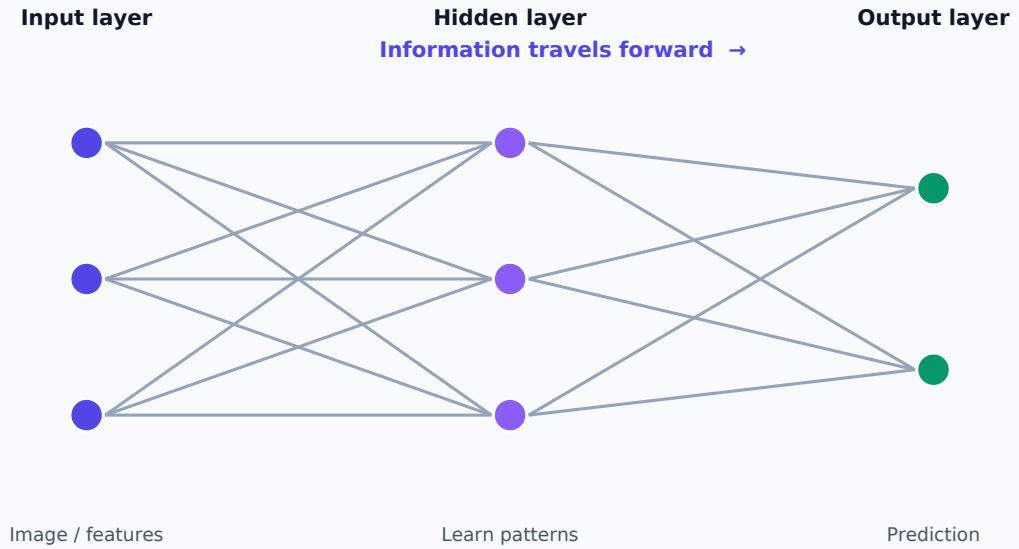
Historical accuracy: Hinton did not receive the Nobel Prize for inventing backpropagation alone. The influential 1986 paper by David Rumelhart, Geoffrey Hinton and Ronald Williams helped establish backpropagation as a practical method for training neural networks.

Key historical milestones

1943	McCulloch and Pitts describe a mathematical model of an artificial neuron.
1958	Frank Rosenblatt introduces the perceptron.
1970	Seppo Linnainmaa publishes an important foundation for reverse-mode automatic differentiation.
1974	Paul Werbos discusses applying backpropagation ideas to neural networks.
1986	Rumelhart, Hinton and Williams publish an influential paper on learning by back-propagating errors.

2. Concepts to understand first

These are the small set of ideas needed to follow the backpropagation process.



Neural network: A computer model made of connected artificial neurons that learns patterns from data.

Input layer: The first layer. It receives the information, such as an image or numbers.

Hidden layer: Layers between input and output that process information and learn useful patterns.

Weights and biases: Weights control the influence of inputs. Biases help shift a neuron's output.

Forward propagation: Information moves from input through the network to produce a prediction.

Supervised learning: A model learns from examples that include the correct answer.

Labeled data: An input paired with its correct answer. Example: a cat image labelled “cat”.

Loss function: A calculation that measures how different a prediction is from the correct answer.

Gradient: A value that describes how the loss changes when a parameter changes.

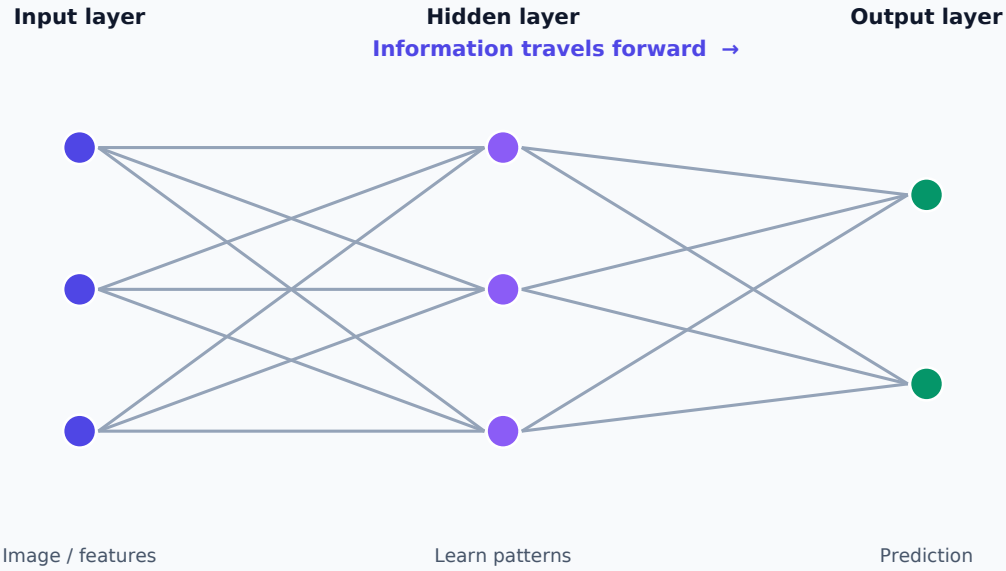
Gradient descent: An optimization method that uses gradients to adjust weights and reduce loss.

Keep this distinction clear

Backpropagation calculates gradients. Gradient descent (or another optimizer) uses those gradients to update the model's weights.

3. Forward propagation: how a prediction is made

Example: an image-classification model receives a picture and predicts what it contains.



The process

- Input: the image is converted into numbers the network can process.
- Processing: information moves through hidden layers. The network responds to patterns it learned during training.
- Output: the final layer produces scores or probabilities for possible classes.

Example prediction

Class	Model score
Cat	80%
Dog	15%
Rabbit	5%

Why did it predict “cat”? During training, it learned patterns from many labeled examples. Its output reflects patterns in the training data; it is not human-like understanding.

4. What happens after the forward pass?

The prediction is compared with the correct label. A loss function measures how far the model's output is from the target.

Example

Correct label: cat
Model output: cat = 0.80
The loss calculation tells the training process how well the model did. A lower loss generally means a closer match to the target.

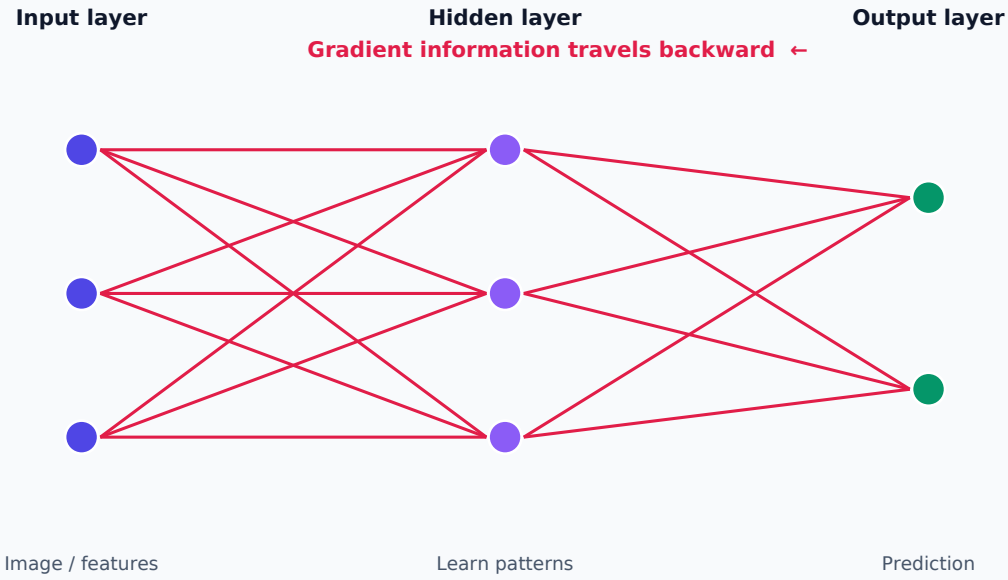
Then developers use a training method to improve the model. They do not normally adjust every weight by hand. Instead, the training algorithm calculates how parameters should change.

The question that leads to backpropagation

If the final prediction is wrong, how can we work out which weights across all the layers contributed to that error? How should each weight change?

Randomly changing weights is inefficient, especially when a network has many layers and a very large number of parameters.

5. The problem backpropagation helps solve



The credit assignment problem

A network's final prediction depends on many connected weights. When the prediction is wrong, we need to estimate how much each weight contributed to the final loss. This is known as the credit assignment problem.

How backpropagation helps

- It calculates gradients that show how the loss changes with each weight.
- It passes gradient information backward through the layers.
- It uses the chain rule to combine the effects of connected operations.
- It provides the information an optimizer needs to update the weights.

The breakthrough

Instead of guessing which weights to change, the network can calculate a direction for changing them so the loss can be reduced.

6. What is backpropagation?

Definition: Backpropagation is an algorithm used to train neural networks. It calculates how much each weight contributes to the prediction loss by propagating gradients backward through the network.

In simple words: it works out how changing a connection's weight would affect the model's error.

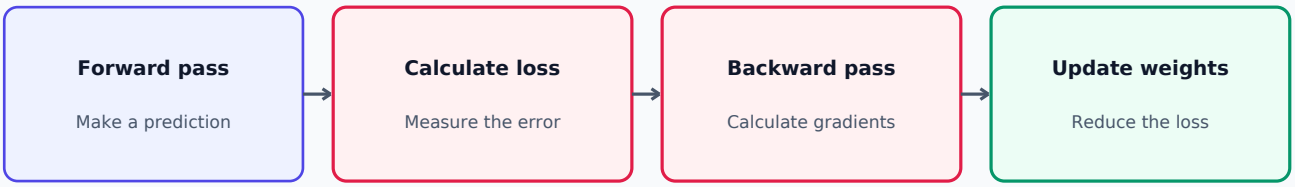
Its main job

- Calculate gradients for weights and biases.
- Send gradient information from the output layer back toward earlier layers.
- Provide gradients to the optimizer for parameter updates.

Backpropagation and gradient descent are not the same

Backpropagation	Gradient descent
Calculates gradients: how the loss changes with parameters.	Uses gradients to update parameters in a direction intended to reduce loss.

7. How backpropagation works: the full cycle



Step-by-step

- 1. Forward pass: send input through the network and produce a prediction.
- 2. Calculate loss: compare the prediction with the correct target.
- 3. Start backward pass: calculate how the loss changes with the output.
- 4. Move backward: use the chain rule to calculate gradients for earlier layers.
- 5. Collect gradients: calculate gradients for weights and biases.
- 6. Update weights: an optimizer such as gradient descent adjusts the parameters.
- 7. Repeat: train on more examples and continue reducing loss.

Basic weight-update equation

New weight = Old weight − Learning rate × Gradient

The learning rate controls the size of each update. The gradient indicates how the loss changes with the weight.

A smaller loss on the next pass suggests that the update helped for that training step. Training repeats over many examples and many steps.

8. What happens internally? The chain rule

A weight affects a neuron's output. That output affects the next layer, and eventually the final loss. The chain rule connects these effects.

Simple chain of operations

Weight → weighted sum → activation → later layers → loss

To find how a weight affects the loss, backpropagation multiplies the relevant derivatives along this path.

Illustrative chain-rule equation

If a weight w affects the loss through z and a, then:

$$\partial L/\partial w = (\partial L/\partial a) \times (\partial a/\partial z) \times (\partial z/\partial w)$$

What the pieces mean

- $\partial L/\partial a$: how the loss changes when the activation changes.
- $\partial a/\partial z$: how the activation changes when its input changes.
- $\partial z/\partial w$: how the weighted sum changes when the weight changes.

Backpropagation reuses intermediate derivative calculations. This is much more efficient than independently recalculating the full effect of every weight.

This is why derivatives, partial derivatives and the chain rule are the most important mathematics to understand before studying backpropagation.

9. Where is backpropagation used?

Backpropagation is used to train many kinds of neural networks. These networks help power products and services people use every day.

Image recognition

CNNs learn visual patterns for face recognition, photo search and object detection.

Medical imaging

Neural networks learn patterns in X-rays, CT scans and MRI images to support medical analysis.

Voice assistants

Neural networks learn patterns in audio and speech.

Translation

Neural networks learn relationships between words and languages.

Self-driving technology

Vision models learn to recognize lanes, vehicles, signs and pedestrians.

Large language models

Transformer networks use backpropagation-based training to adjust model parameters.

Important distinction: backpropagation is mainly part of model training. When a trained model is being used to make a prediction (inference), the usual process does not need to calculate training gradients or update weights.

10. Advantages, disadvantages and limitations

Advantages

- Learns useful patterns from examples instead of requiring every feature to be hand-coded.
- Calculates gradients efficiently across many connected layers.
- Supports the training of deep neural networks.
- Works with many architectures and optimization methods.

Disadvantages

- Large models can require substantial computing power and time.
- Training often requires significant memory to store intermediate values.
- Results depend on data quality, model design and training settings.

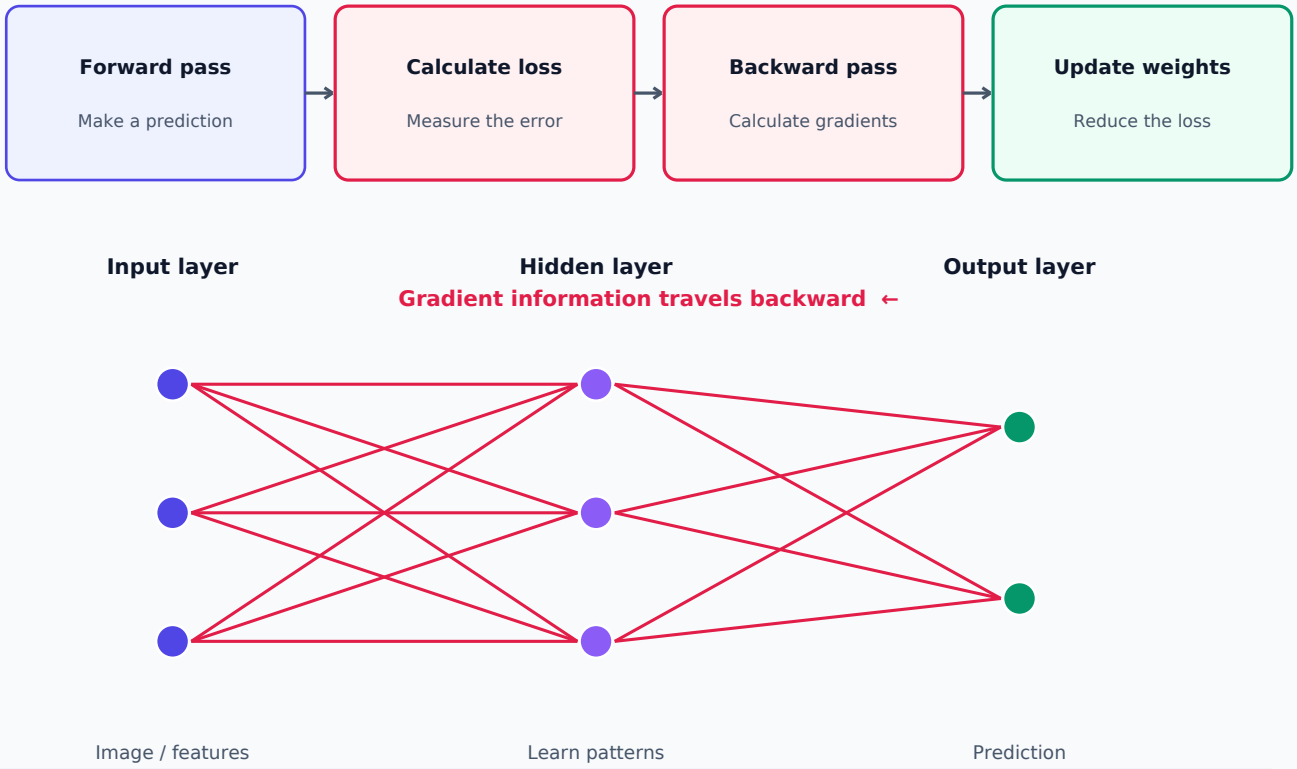
Limitations and challenges

Vanishing gradients	Gradients become very small, so earlier layers may learn slowly.
Exploding gradients	Gradients become very large and can make training unstable.
Optimization difficulties	Training can encounter saddle points or difficult loss landscapes.
Differentiability	Standard backpropagation needs differentiable operations along the path being trained.
No global guarantee	It does not guarantee finding the globally best possible weights.
Biological difference	It is not a close model of how biological brains learn.

Conclusion

Backpropagation gave neural networks an efficient way to calculate how their weights contribute to prediction error. Together with an optimizer such as gradient descent, it helps neural networks learn from examples and powers much of modern deep learning.

Quick revision: the whole idea in one page



Remember these three points

- Forward propagation makes a prediction.
- Backpropagation calculates gradients by moving backward through the network.
- Gradient descent uses those gradients to update weights.

Suggested final line for your presentation

“Backpropagation does not make a neural network intelligent by itself. It gives the network a practical way to learn from its mistakes by calculating how its parameters should change.”

References and further reading

Use these sources for the historical claims and for deeper study. The explanations in this guide are simplified for a classroom presentation.

- Nobel Prize (2024), The Nobel Prize in Physics 2024 — John Hopfield and Geoffrey Hinton. https://www.nobelprize.org/prizes/physics/2024/summary/
- Rumelhart, D. E., Hinton, G. E., & Williams, R. J. (1986). Learning representations by back-propagating errors. Nature, 323, 533–536. https://doi.org/10.1038/323533a0
- Werbos, P. J. (1974). Beyond Regression: New Tools for Prediction and Analysis in the Behavioral Sciences. Harvard University doctoral thesis.
- Linnainmaa, S. (1970). The representation of the cumulative rounding error of an algorithm as a Taylor expansion of the local rounding errors. Master's thesis, University of Helsinki.

Presentation reminder

Spend the most time explaining the prediction, loss, credit assignment problem, backward gradient calculation and weight update. Those steps tell the complete story of why backpropagation matters.